

6	A steel ball of mass m is dropped vertically onto a steel table. The ball is in contact with the table for a very short duration and it rebounds with very little loss in kinetic energy. Which one of the following statements concerning the average force F exerted on the ball by the table during the collision is correct?	
	A	F is smaller than mg .
	B	F is greater than mg .
	C	F is greater than the force that the ball exerted on the table during the collision.
	D	F is smaller than the force that the ball exerted on the table during the collision.

Ans: B

Upwards contact force on the ball must be larger than weight otherwise you can't get a net force upwards to cause the ball to rebound

Options C and D are wrong because the force exerted by the ball on the table is always equal to F using Newton's 3rd law.

[Turn over

